

Chirping Laser Pulses for Spectral-Focusing CARS

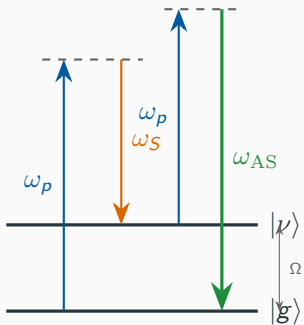
Dispersion, Resolution, and Signal Trade-offs

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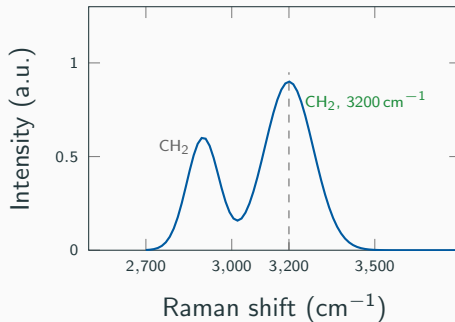
Maîtrise en biophotonique

CARS detects lipid bond vibrations coherently, without labels



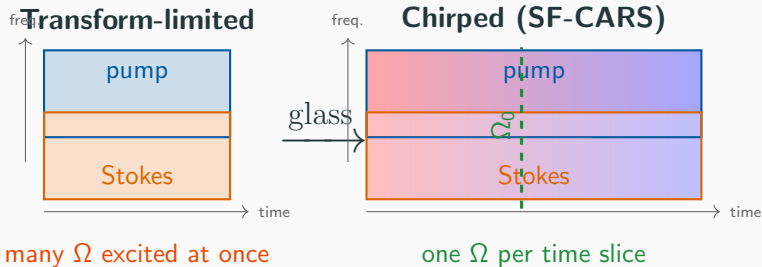
$$\omega_{AS} = 2\omega_p - \omega_s$$

resonance: $\Omega = \omega_p - \omega_s$

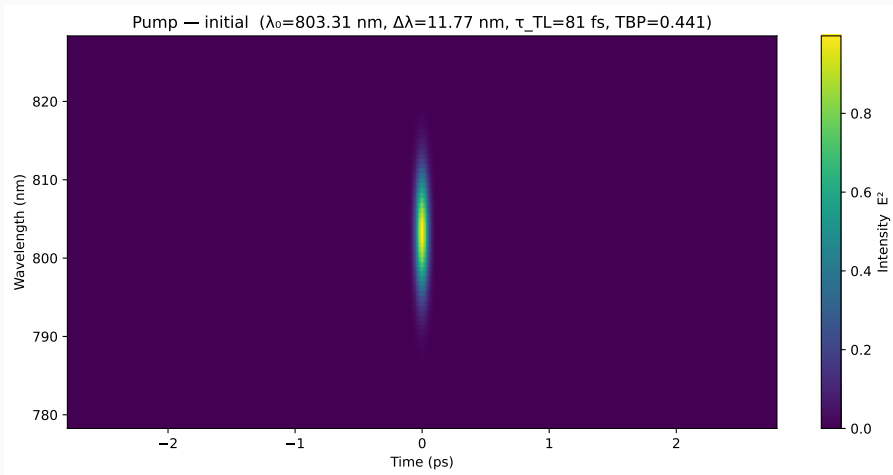


Pump: 803 nm Stokes: 1041 nm AS:
654 nm

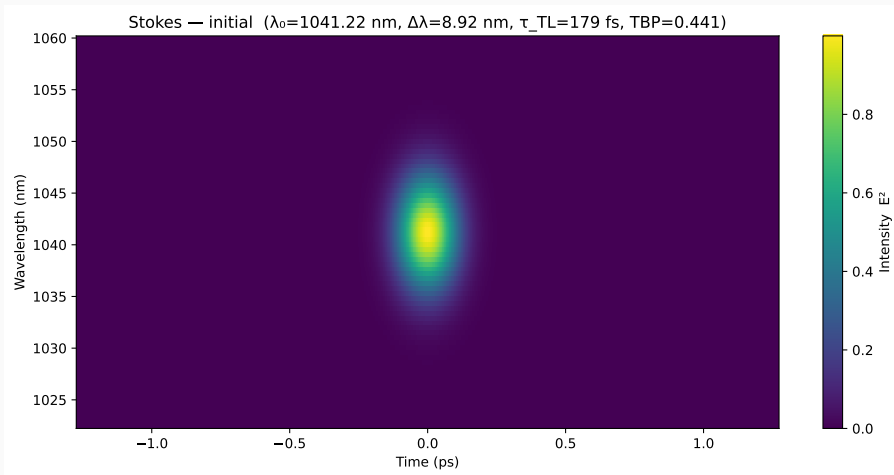
Stretching both pulses with glass makes the instantaneous frequency difference constant



The pump starts transform-limited: all 80 spectral slices peak together at $t = 0$

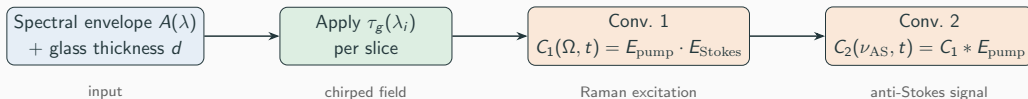


The Stokes pulse is transform-limited too, but twice as long at 1041 nm



The simulation shifts each spectral slice by its group delay, then computes two convolutions

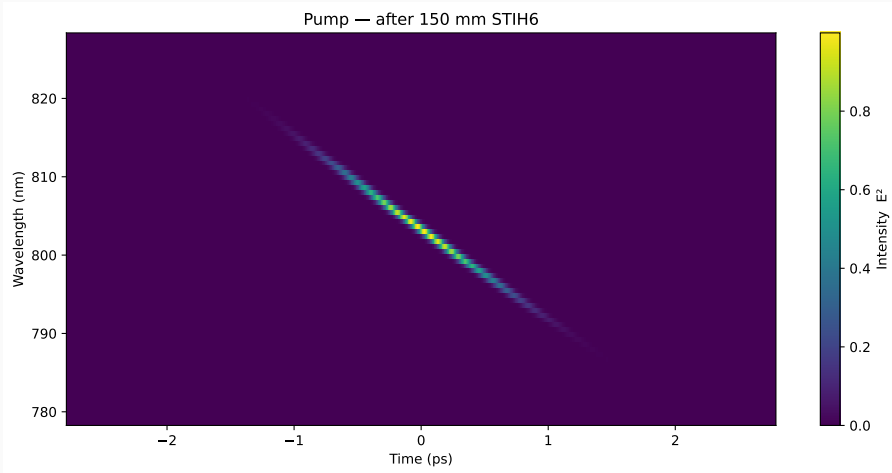
$$\tau_g(\lambda_i) = \frac{d [n_g(\lambda_i) - n_g(\lambda_0)]}{c}, \quad n_g = n - \lambda \frac{dn}{d\lambda}$$



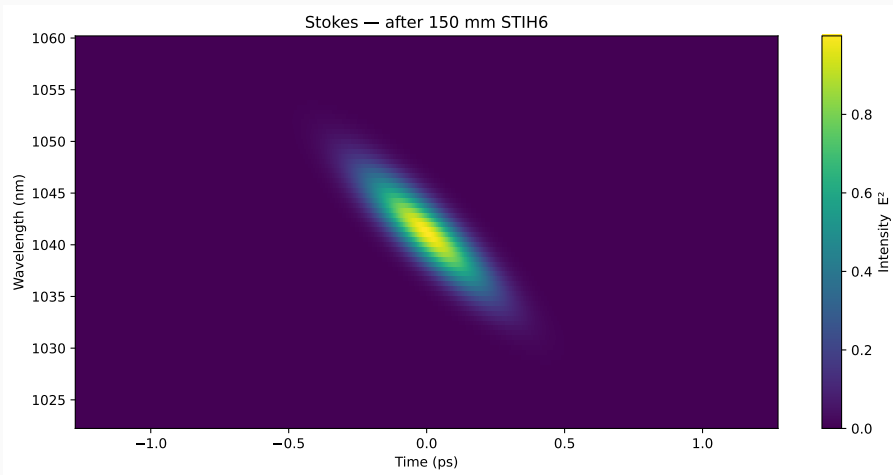
All operations on **field amplitudes** (E), not intensities — CARS coherence scales as

$$E_{\text{pump}} \cdot E_{\text{Stokes}}.$$

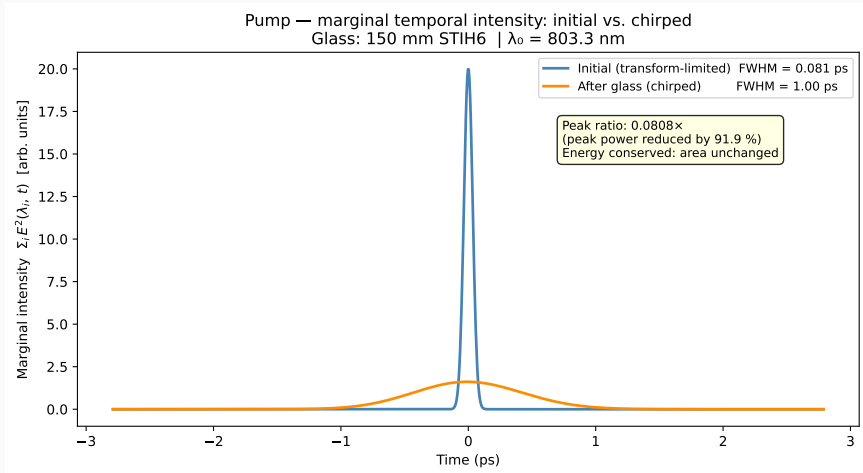
15 cm of S-TIH6 stretches the pump 52 $\times$, mapping frequency onto time



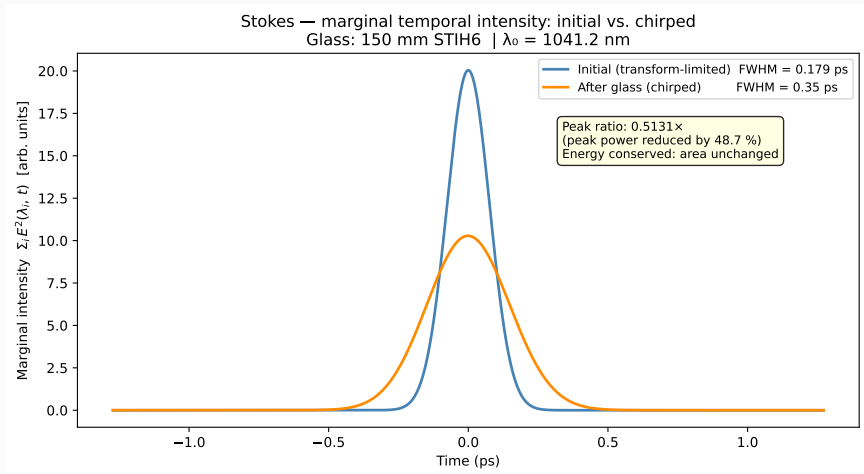
The same glass chirps the Stokes only $7\times$: dispersion is weaker at 1041 nm



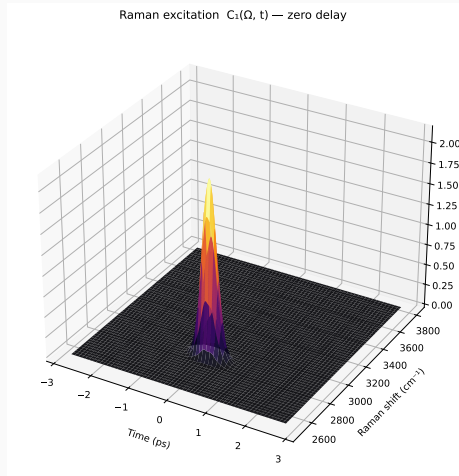
Chirping costs 92% of pump peak power — but total energy is conserved



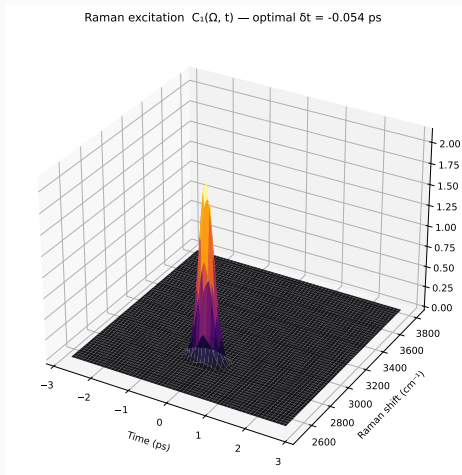
Stokes loses only 49% of peak power: it sweeps frequency more slowly



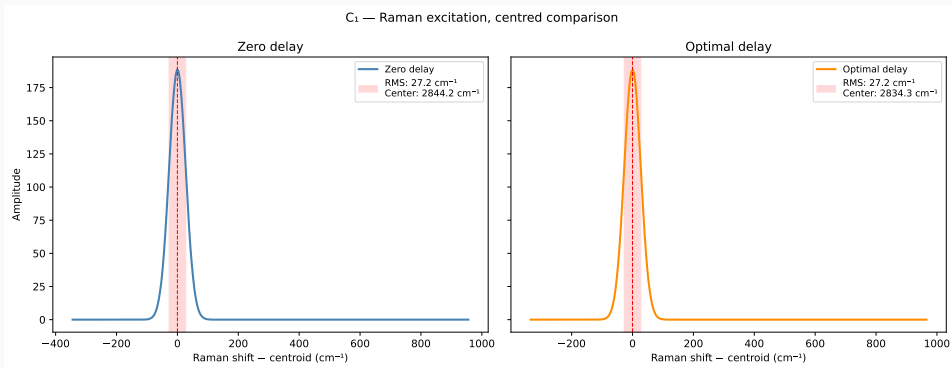
At each instant a single Raman mode is driven: the horizontal ridge is spectral focusing



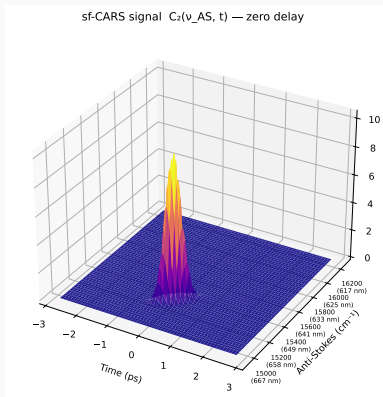
Optimal delay is zero: matched chirp rates naturally align the two beams



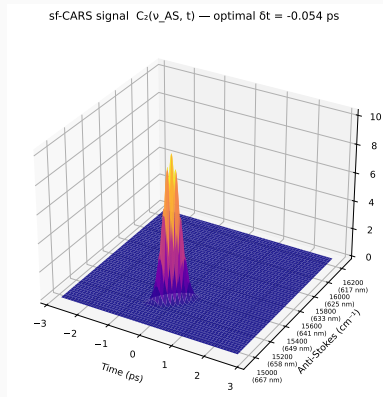
Matched chirp delivers 27.1 cm^{-1} Raman resolution — and timing barely matters



The probe pump converts Raman coherence into anti-Stokes light at 654 nm

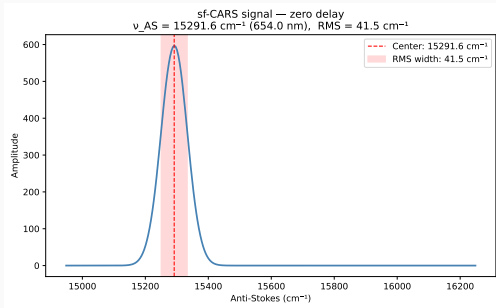


Zero delay

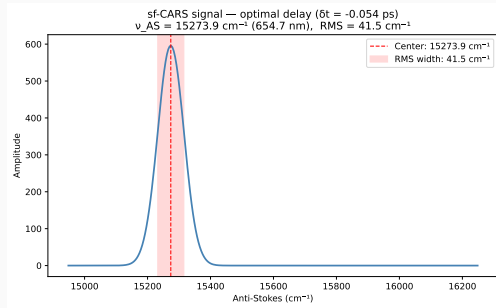


Optimal delay ($\delta t^* \approx 0$)

Both delays yield the same 41.5 cm^{-1} anti-Stokes linewidth

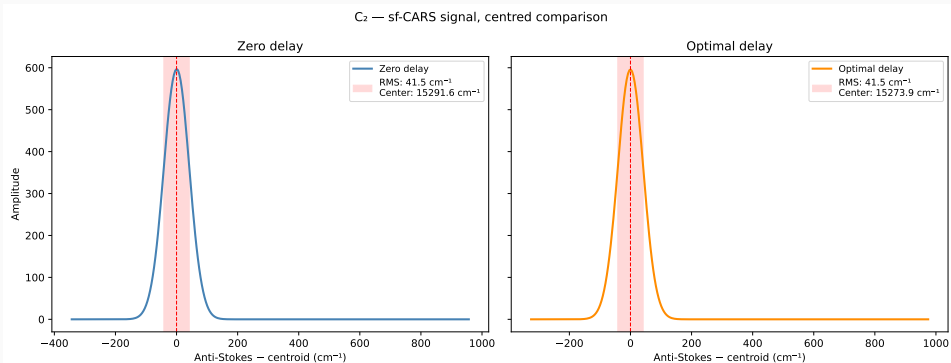


Zero delay

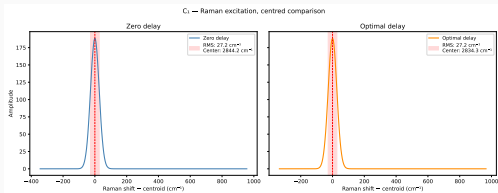


Optimal delay — **RMS = 41.5 cm^{-1}**

The degenerate probe broadens the anti-Stokes spectrum 53% beyond C_1

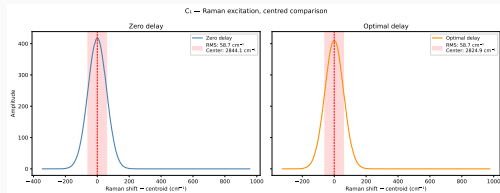


Halving pump glass from 15 to 5 cm doubles the Raman excitation linewidth



Scenario A — pump 15 cm

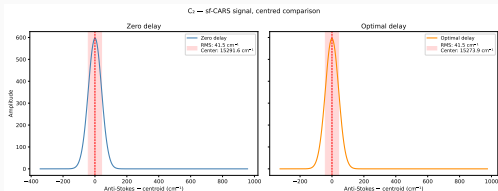
$$\text{RMS} = 27.1 \text{ cm}^{-1}$$



Scenario B — pump 5 cm

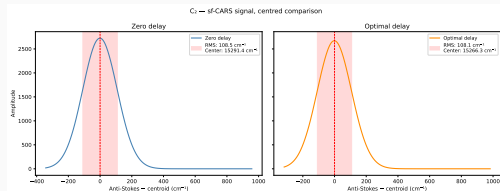
$$\text{RMS} = 58.7 \text{ cm}^{-1}$$

Mismatched chirp degrades anti-Stokes resolution $2.6\times$



Scenario A — pump 15 cm

$$\text{RMS} = 41.5 \text{ cm}^{-1}$$



Scenario B — pump 5 cm

$$\text{RMS} = 108.5 \text{ cm}^{-1}$$

More pump glass: $2.6\times$ better resolution, $3\times$ lower peak power — the same knob controls both

	Scenario A pump 15 cm	Scenario B pump 5 cm
Pump GD spread	4.21 ps	1.40 ps
Pump elongation	$\approx 52\times$	$\approx 17\times$
Pump peak ratio	0.080 (−92%)	0.22 (−78%)
C_1 RMS (cm^{-1})	27.1	58.7
C_2 RMS (cm^{-1})	41.5	108.5
Optimal δt	≈ 0	$\neq 0$

Chirp matching ($2\times$ glass ratio) gives $2.2\times$ better C_1 resolution.

Anti-Stokes (C_2) improves $2.6\times$ — only Stokes glass is fixed.

More glass $\Rightarrow$ narrower linewidth *and* lower signal.

Spectral focusing turns femtosecond bandwidth into narrowband Raman sensitivity — chirp matching is the key

1. **S-TIH6 imprints a linear frequency-to-time map.** Pump: 80 fs $\rightarrow$ 4.2 ps (52 $\times$); Stokes: 179 fs $\rightarrow$ 1.3 ps (7 $\times$).
2. **Peak power drops but energy is conserved.** Pump peak ratio = 0.080 = $1/n_{\text{eff}}$; $\int I dt$ ratio = 1.000.
3. **Matched chirp rates achieve 27 cm⁻¹ Raman resolution** from an 11.77 nm input bandwidth — the time-frequency mapping acts as a narrowband filter.
4. **Mismatch is costly.** Pump 5 vs. 15 cm: C_1 degrades 2.2 $\times$, C_2 degrades 2.6 $\times$.
5. **The degenerate probe adds 53% spectral width.** C_2 RMS = 41.5 cm⁻¹ > C_1 RMS = 27.1 cm⁻¹; a narrowband probe would sharpen the final spectrum.

Thank you

Questions?

Code: `branch 3d-propagation — resolution.py`

All plots and simulation scripts available in the repository.